Nonparametric Hypothesis Tests Last-Minute Cheatsheet

What Are Nonparametric Methods?

Key Idea	Nonparametric tests do **not** require any distributional assumptions about the population. They rely solely on the observed data, not on an assumed probability model.
Contrast	Parametric methods (t-test, ANOVA, F-test) depend on both the data *and* an assumed distribution. Nonparametric methods depend on **only the data**.
When to Use	Small samples, ordinal data, non-normal populations, or when distributional assumptions cannot be verified.

Parametric vs. Nonparametric Estimation

Given a random sample $T_1, T_2, \ldots, T_n$ (e.g., battery lifetimes):

PARAMETRIC

ASSUME EXPONENTIAL(Λ)

$$F(x, \lambda) = 1 - e^{-\lambda x}$$

MLE OF Λ

$$\hat{\lambda} = \frac{\sum_{i=1}^{n} t_i}{n}$$

ESTIMATED CDF

$$\hat{F}(x) = 1 - e^{-\hat{\lambda} x}$$

NONPARAMETRIC

EMPIRICAL CDF (NO MODEL)

$$\hat{F}(x) = \frac{r}{n}$$

No parameters to estimate. The CDF is built directly by counting observations $\leq x$. Simple, robust, model-free.

EXAMPLE

$n = 20$ batteries, 6 have lifetime ≤ 9 hrs $\Rightarrow \hat{F}(9) = 6/20 = 0.30$.

The Sign Test — Hypothesis & Setup

Let $X_1, \ldots, X_n$ be a random sample from a **continuous** distribution F. Test whether the **median** m equals m_0:

HYPOTHESES (TWO-SIDED)

$$H_0 : \; m = m_0 \qquad \text{vs} \qquad H_1 : \; m \neq m_0$$

Why the median? For any continuous F, the median satisfies $P(X \leq m) = 1/2$. Under H_0, half the observations are expected to fall below m_0.

Sign Test — Test Statistic

INDICATOR VARIABLE

$$I_i = \begin{cases} 1 & \text{if } X_i \leq m_0 \\ 0 & \text{otherwise} \end{cases}$$

Each observation reduces to a **sign**: is it $\leq m_0$ or not? Magnitudes are discarded — hence **Sign Test**.

TEST STATISTIC

$$v = \sum_{i=1}^{n} I_i$$

DISTRIBUTION UNDER H_0

$$v \sim B\left(n, \tfrac{1}{2}\right)$$

Under H_0, each X_i has $P(X_i \leq m_0) = 1/2$, so v follows **Binomial** $(n, 1/2)$.

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P-Value Formula | Simplified P-Value | Worked Example | Extensions

Sign Test — P-Value Calculation

GENERAL TWO-SIDED P-VALUE

$$p = 2\,\min\left\{P\big(B(n,p) \le v\big),\ \ P\big(B(n,p) \ge v\big)\right\}$$

Intuition: Take the tail probability in the "worse" direction and double it. If v is far from $n/2$, the p-value is small → evidence against H_0.

Simplification using symmetry of $B(n,1/2)$:

Since $P(B(n,p) \ge v) = P(n - B(n,p) \le n - v) = P(B(n,1-p) \le n - v)$:

SIMPLIFIED P-VALUE (P = 1/2)

$$p = \begin{cases} 2\,P\big(B\big(n,\tfrac{1}{2}\big) \le v\big) & \text{if } v \le n/2 \\ 2\,P\big(B\big(n,\tfrac{1}{2}\big) \le n - v\big) & \text{if } v \ge n/2 \end{cases}$$

Rule: Always use the **smaller** of v and $n - v$ as the argument. This avoids computing the upper tail directly.

Worked Example (Ex 12.2a)

1. **Given:** $n = 200$, $v = 120$ values below m_0, 80 above. Test $H_0 : m = m_0$ vs $H_1 : m \neq m_0$.
2. **Identify:** $v = 120 > n/2 = 100$, so use the second case.
3. **Compute** $n - v = 200 - 120 = 80$.
4. **P-value:** $p = 2\,P(B(200,1/2) \le 80)$

$$p = 2\,P\big(B(200,\tfrac{1}{2}) \le 80\big)$$

Decision: If $p < \alpha$, reject H_0 (median $\neq m_0$). If $p \ge \alpha$, fail to reject H_0.

Extensions of the Sign Test

OTHER QUANTILES

The Sign Test can test **any quantile** q (e.g., Q_1, Q_3), not just the median.

GENERAL QUANTILE TEST

$$H_0 : q = q_0 \ \Rightarrow\ v \sim B(n, p_0)$$

p_0 = theoretical proportion below the quantile ($p_0 = 0.25$ for Q_1, 0.75 for Q_3). P-value uses $B(n,p_0)$ instead of $B(n,1/2)$.

PAIRED DATA (VS PAIRED T-TEST)

For paired (X_i, Y_i), define $D_i = X_i - Y_i$. Test whether the **median of differences** is zero.

PAIRED HYPOTHESIS

$$H_0 : m_D = 0$$

$I_i = 1$ if $D_i > 0$, else 0. Then $v = \sum I_i \sim B(n,1/2)$ under H_0. Nonparametric alternative to the paired t-test when normality cannot be assumed.

Quick Decision Rules

SCENARIO	TEST STATISTIC	DECISION RULE
MEDIAN TEST $H_0 : m = m_0$	$v = \#\{X_i \le m_0\}$ $v \sim B(n,1/2)$	Reject H_0 if $p = 2\min\{P(B \le v),\ P(B \ge v)\} < \alpha$
QUANTILE TEST $H_0 : q = q_0$	$v = \#\{X_i \le q_0\}$ $v \sim B(n,p_0)$	Reject H_0 if $p = 2\min\{P(B \le v),\ P(B \ge v)\} < \alpha$
PAIRED TEST $H_0 : m_D = 0$	$v = \#\{D_i > 0\}$ $v \sim B(n,1/2)$	Reject H_0 if $p = 2\min\{P(B \le v),\ P(B \ge v)\} < \alpha$

Pro tip: For large n, use the Normal approximation $B(n,1/2) \approx N(n/2,\ n/4)$ to compute p-values without binomial tables.